

Gokulraj M

Software Engineer · AI Engineer · Google Cloud / Looker Developer

Tiruvannamalai, Tamil Nadu | +91-7695866703 | gokulraj.gokul3003@gmail.com | linkedin.com/in/gokul-raj3003 | Portfolio: gokul.quest

Software Engineer building production AI infrastructure, provider-agnostic LLM platforms, and scalable evaluation systems alongside Google Cloud and Looker analytics at TCS. Independently architected and deployed Brok's Forge, an AI Engineering Platform for evaluating AI agents, and Kuriosity, a production RAG learning platform. Experienced in LLM integration, provider abstraction, RAG pipelines, and multi-tenant backend architecture.

TECHNICAL SKILLS

AI / LLM Engineering	AI agent evaluation & benchmarking, provider-agnostic model abstraction, RAG pipelines, prompt engineering & versioning, vector databases (ChromaDB), embeddings (Cohere)
Backend & Data	Java, Spring Boot, Spring Security/Data JPA, Node.js, Express, REST APIs, JWT auth, AES-256-GCM encryption, multi-tenant architecture, PostgreSQL, Redis, Firebase
Cloud & BI	Google Cloud Platform (GCP), Looker, LookML, BigQuery, Cloud Storage, Dataflow
DevOps & Infrastructure	Docker, Docker Compose, AWS EC2, Nginx, GitHub Actions CI/CD, Railway, Vercel, Let's Encrypt/TLS
Frontend & Languages	React, Next.js, Vite, TypeScript, TailwindCSS, Java, Python, JavaScript, C#, C++, SQL
Tools & Concepts	Git/GitHub, Postman, JUnit, OOP, Data Structures & Algorithms, System Design

EXPERIENCE

Tata Consultancy Services (TCS)

Dec 2025 – Present

Software Engineer, Google Cloud / Looker Developer

Chennai, Tamil Nadu

- Design and develop production-grade LookML models, views, and explores on Google Cloud Looker, enabling secure self-service analytics for Finance, Operations, and Marketing stakeholders.
- Restructured BigQuery SQL queries and aggregation logic, reducing average dashboard load time by ~35% across 10+ production Looker dashboards.
- Implemented row-level security and access-group policies in LookML to enforce multi-tenant data governance.

NIT Trichy

Sep 2024 – Oct 2024

Game Developer Intern

Trichy, Tamil Nadu

- Owned the end-to-end lifecycle of a game on online-bullying awareness, from design through deployment, using Unity and C#; optimized frame-rate and game-loop mechanics.

MIC Chennai

Aug 2023

VR Developer Intern

Chennai, Tamil Nadu

- Built immersive VR experiences with Unity's XR Interaction Toolkit and C#, improving simulation fidelity through iterative performance testing.

PROJECTS

Brok's Forge — AI Engineering Platform

broksforge.gokul.quest

Java 21, Spring Boot 3, PostgreSQL, Redis, Docker, AWS EC2, Nginx, Next.js, TypeScript, GitHub Actions CI/CD

- Architected and built a self-hosted AI engineering platform for evaluating and benchmarking AI agents across frameworks including Spring AI, LangGraph, CrewAI, and AutoGen through a provider-agnostic invocation layer independent of any single LLM vendor.
- Designed a model-invocation abstraction layer with adapters for OpenAI, Anthropic, Groq, Gemini, OpenRouter, and DeepSeek, decoupling the evaluation engine from any one provider SDK.
- Engineered a three-level evaluation pipeline (Job → Run → Result) that completes outbound calls before opening a database transaction to persist results, avoiding connection-pool exhaustion under slow external calls.
- Designed immutable, versioned datasets/prompts for reproducible results, and a multi-tenant architecture with compound-key tenant isolation plus AES-256-GCM encrypted credential storage; deployed on AWS EC2 (Docker, Nginx, TLS) with automated GitHub Actions CI/CD.

Kuriosity — AI-Powered Student Learning Platform

kuriosity.gokul.quest

React, Vite, Express, PostgreSQL, ChromaDB, Groq (LLM), Cohere (Embeddings), Resend, Railway, Vercel

- Architected and shipped a full-stack RAG application end-to-end: users upload study material and an AI student ("Kurio") asks context-grounded questions, evaluates answers, and tracks topic mastery across sessions.
- Built a custom RAG pipeline (PDF/PPT parsing, chunking, Cohere embeddings, ChromaDB similarity search) to ground every response in source material rather than the LLM's general knowledge.
- Engineered an evaluation layer scoring free-text answers against retrieved chunks (structured JSON: accuracy, missing concepts, revision topics) and a PostgreSQL-backed cross-session topic-memory system; deployed a three-platform stack (Vercel, Railway) with JWT + OTP auth and rate limiting.

Violence Detection & Alert System

YOLOv6, MobileNetV2, Python

- Built a real-time computer vision pipeline that detects violent activity in video streams and automatically triggers safety alerts.

EDUCATION

SRM Institute of Science & Technology

Trichy, Tamil Nadu

B.Tech – Computer Science Engineering, CGPA: 7.83

May 2021 – Jul 2025

CERTIFICATIONS

- Associate Cloud Engineer (Google certified)
- HackerRank — Java (Certified)
- CodeChef — Python
- AR/VR Game Development — CDAC Pune

PROFILE LINKS

GitHub: github.com/gokulraj9488

Portfolio: gokul.quest

LinkedIn: linkedin.com/in/gokul-raj3003